

When Relevance Does Not Transfer

Consumer-Dependent Retrieval Utility in Sequential Decision Systems

Anonymous Author(s)

Anonymous Institution

Abstract. Retrieval is evaluated by relevance, and increasingly by the quality of the text a language model then generates. A growing class of deployments instead feeds retrieved evidence to a decision policy, where an error’s cost depends on which action it induces and on how the system evolves afterwards. We build an executable benchmark for that setting — an operational report feed, a retrieval step, a fixed probabilistic interpreter, a fixed controller, and a simulator that returns realised utility — and report a negative result. The value of a ranking is not a property of the ranking. At a matched evidence budget, every ranking system we test *helps* one downstream consumer and *harms* another: the retriever $\times$ consumer interaction changes sign for all four, and ranking quality orders systems correctly for one consumer (rank-reversal rate 0.00) but not the other (0.33). The information is present and extractable — an oracle relevance selector helps both consumers — so the failure is a mismatch between retrieval and its consumer, not an absence of useful evidence. Supporting diagnostics locate the mismatch: text-free, shuffled-text and label-only controls show that belief accuracy and probabilistic sophistication do not imply operational value, and a perfect-semantic belief reference is not an upper bound on realised return. We release the benchmark, its frozen artifacts and its controls; every table regenerates from the stored episodes.

Keywords: Retrieval evaluation · Downstream utility · Evidence selection · Consumer dependence · Calibration.

1 Introduction

An operator monitoring a supply network receives a stream of reports. Most concern other sites; some are stale; a few describe conditions that should change what they order this period. Choosing which reports to act on is an information access problem, and its consequences are sequential: an order placed now arrives several periods later, into a state that earlier decisions produced.

Standard retrieval evaluation scores such a system by topical relevance, and downstream-aware evaluation such as eRAG scores it by the quality of the text a model subsequently generates [18]. Both treat the value of a ranking as a property of the ranking. This paper reports that in a consequential setting it is not.

We ask when retrieved text improves sequential decisions, and make the whole path executable so the question can be answered rather than argued:

$$q, \mathcal{D} \longrightarrow R_k(q, \mathcal{D}) \longrightarrow b(z \mid \cdot) \longrightarrow \pi(a_t \mid s_t, b) \longrightarrow s_{t+1} \longrightarrow U. \quad (1)$$

A retrieval system R_k selects k documents from a candidate pool $\mathcal{D}$ under a fixed query q ; a fixed interpreter maps the selected text to a belief b over operational regimes; a fixed controller maps state and belief to an action; a simulator returns realised utility U . Holding everything after retrieval constant makes differences in U attributable to evidence selection. Varying only the interpreter, with retrieval held constant, isolates the consumer.

The result. At $k = 3$, every ranking system — random, BM25, TF-IDF cosine and dense retrieval — produces a positive paired effect for a rule-based consumer and a negative one for a calibrated classifier consumer. The retriever $\times$ consumer interaction changes sign in all four cases and reaches +89 reward units. Ranking quality orders systems exactly as realised utility does for the first consumer and disagrees on a third of pairs for the second. The evidence itself is not the problem: an oracle relevance selector improves *both* consumers, by +60.6 and +69.0 respectively. What fails is the match between what practical rankers retrieve and what a particular consumer does with it.

Contributions. (i) An executable benchmark that scores evidence selection by the sequential decision utility it induces, with retrieval quality reported on the same episodes (Section 4). (ii) A measured negative result — consumer-dependent retrieval utility — reported as an interaction, a rank-reversal rate, episode-level correlations and a harmful-retrieval rate rather than a single headline number. (iii) Diagnostics that locate the mismatch: text-free, shuffled-text and label-only controls showing that belief accuracy and probability quality do not imply operational value (Section 5). (iv) A reproducible artifact with the estimand, weighting and uncertainty declared, in which every table in this paper regenerates from the stored episodes.

Scope. The report feed and environments are synthetic. We introduce no retrieval model and claim no interpreter or controller is preferable in general. The claim is about *evaluation*: a retrieval system that feeds a decision policy cannot be scored independently of that policy, and the controls needed to support such a claim are stronger than those usually reported.

2 Related work

Downstream-aware retrieval evaluation. eRAG evaluates retrieval by the downstream performance of the language model that consumes it, showing that upstream relevance can be a poor proxy for end-task utility [18]. CUE-R estimates per-evidence utility by intervention and includes no-retrieval controls;

GroGU argues grounding utility is consumer-specific. Both are recent preprints and we treat them as such [3,8]. Our downstream consumer is a control policy in a stochastic environment rather than a generator, so the estimated quantity is a trajectory’s realised return, not text quality, and the error asymmetry comes from the environment’s cost structure rather than a judge. Where GroGU argues for consumer specificity, we measure it end to end and find it strong enough to reverse the sign of a ranking’s value. That relevance is situational and task-dependent long predates this work [2]; the contribution here is an instrument for one consequential task, not the observation that consequences matter.

Decision-focused prediction and calibration. Predict-then-optimize and decision-focused learning train or evaluate predictions through a downstream objective [5,6,14]; value-aware model learning makes the analogous point for learned dynamics [7,13], decision calibration formalises dependence on a decision-maker [19], and value-of-information theory gives the classical account of when information changes a decision [10]. Probability quality is standardly assessed with proper scores and calibration diagnostics [9,4,15]. We do not learn through the objective: we hold the pipeline fixed and vary what is supplied to it, which makes this an evaluation rather than a training method.

Environment and baselines. The multi-echelon simulator derives from an inventory benchmark lineage in operations research [11] and is used at a pinned published version [1]. Retrieval baselines are Okapi BM25 [17] and cosine similarity over sentence embeddings [16], scored with nDCG [12].

3 Framework

3.1 Objects and estimand

Let z be a latent operational regime, $\mathcal{D}$ a candidate pool of reports and q a fixed query expressing a standing information need. A retrieval system returns an ordered list; its top k documents are the evidence. A fixed interpreter maps their concatenation to a belief $b(z)$, and a fixed controller maps state and belief to an action.

Write $G_i(m)$ for the *realised* return of episode i under information source m , $G_i(m) = \sum_{t=0}^{T-1} r(s_t^{(i)}, a_t^{(i)}, z^{(i)})$, undiscounted: the horizon is finite and fixed at $T = 40$, and any discounting inside the environment is part of r . We keep realised returns distinct from their expectation, which we estimate by the weighted benchmark return

$$J_w(m) = \sum_r w_r \operatorname{mean}_{f \in \mathcal{F}_r} \operatorname{mean}_{v \in \mathcal{V}_{r,f}} \operatorname{mean}_{i \in \mathcal{S}} G_i(m; r, f, v), \quad (2)$$

averaging seeds within a template variant v , variants within an ambiguity family f , families within a regime r , and combining regimes with declared weights w_r . The *balanced* estimand uses uniform w_r and is primary; a deployment-prior

estimand is secondary. Group sizes are unequal, so (2) is not an equal-row mean over episodes, and the difference is numerically material.

Effects are reported as paired differences ΔJ against a declared reference: for retrieval, a control that retrieves nothing; for interpreters, the uninformed prior. Where a normalised diagnostic is used, its numerator and denominator carry identical weights. We report it sparingly: the *perfect-semantic reference* — a correct belief passed through the fixed controller — is demonstrably not an upper bound on realised return (Section 5), so normalising by it does not yield a recovered fraction of anything.

3.2 Uncertainty

The design crosses worlds (seeds) with language (templates): every template within a regime is evaluated on the same seeds, so seeds are not independent draws nested under templates. We resample the two axes independently — one *shared* seed sample per regime, reused by every family and variant, plus families and variants with replacement. Intervals are percentile intervals from B replicates; p -values are centred bootstrap test inversions, $(1 + \#\{|b^* - \hat{\theta}| \geq |\hat{\theta}|\})/(B + 1)$, bounded below by $1/(B + 1)$ and never reported as exactly zero. With three designed ambiguity families per regime the family axis is small, so intervals are conditional on the fixed template set and generalisation to new language remains open.

4 Experiment R: consumer-dependent retrieval utility

4.1 Setup

Each episode is a (seed, regime) pair. The candidate pool holds four kinds of document, all in the same operational register: *relevant*, a current report about the operator’s own node; *wrong entity*, a current report about a different node, drawn across all regimes so that some of these errors actively mislead and some are accidentally correct; *stale*, the operator’s node but an event already marked resolved; and *off-topic*, routine corporate traffic attributed to the operator’s node so that it cannot be filtered on entity alone.

Relevance is determined by the query — entity and recency — and never by regime; identifying the regime is the interpreter’s job. That separation is what makes the retrieval task solvable and the dissociation measurable.

Systems: random (floor), BM25, TF-IDF cosine, dense sentence-embedding cosine, and an oracle relevance selector; a no-retrieval control supplies the paired reference. Evidence budgets $k \in \{1, 3, 5\}$ are matched across systems. The two consumers are a rule-based extractor and a calibrated TF-IDF classifier; the controller is a receding-horizon optimiser and is identical throughout.

Table 1. Experiment R at a matched evidence budget $k = 3$. The same rankings, read by two different downstream consumers. Ranking quality is a property of the retrieval alone; ΔJ is the paired effect on realised return against a control that retrieves nothing, with crossed bootstrap intervals. Every ranking system helps the rule-based consumer and harms the calibrated one. ‘Harmful’ is the fraction of retrieved documents that are non-relevant *and* describe a regime other than the true one.

System	Ranking quality		RuleBased		Calibrated TF-IDF	
	nDCG@3	Harmful	ΔJ	95% CI	ΔJ	95% CI
<i>No retrieval</i>	0.000	0.00	+0.0	[+0, +0]	+0.0	[+0, +0]
Random	0.090	0.44	+11.6	[-11, +35]	-49.8	[-78, -19]
BM25	0.274	0.43	+14.9	[+5, +26]	-74.3	[-99, -48]
TF-IDF cosine	0.336	0.32	+25.8	[+14, +39]	-61.1	[-88, -32]
Dense (MiniLM)	0.617	0.29	+48.4	[+32, +65]	-1.5	[-31, +29]
<i>Oracle relevance selector</i>	1.000	0.22	+60.6	[+48, +73]	+69.0	[+40, +96]

4.2 The retrieval task is solvable, and the evidence is there

Ranking quality orders the systems random < BM25 < TF-IDF cosine < dense < oracle, with a wide spread in nDCG@3 (Table 1), and the rate at which retrieved documents actively mislead falls monotonically along the same ordering, from 0.44 for random to 0.22 for the oracle selector. The oracle relevance selector improves *both* consumers, by +60.6 [+48, +73] and +69.0 [+40, +96]. Selecting the right evidence from this pool is therefore both possible and worth doing, for either consumer.

4.3 But its value is not a property of the ranking

Every practical ranking system helps the rule-based consumer and harms the calibrated one (Figure 1). Dense retrieval is +48.4 [+32, +65] for the first and -1.5 [-31, +29] for the second; BM25 is +14.9 [+5, +26] and -74.3 [-99, -48]. The retriever $\times$ consumer interaction changes sign for all four systems and ranges from +49.9 to +89.2 reward units (Table 2). There is no single number that describes what BM25 is worth here.

Ranking quality inherits that ambiguity. For the rule-based consumer the nDCG ordering and the utility ordering agree on every pair (rank-reversal rate 0.00, $\tau_b = +1.00$); for the calibrated consumer they disagree on a third of pairs (0.33, $\tau_b = +0.33$). An evaluator who selected a retriever on nDCG would be right every time for one consumer and wrong a third of the time for the other, with no upstream signal distinguishing the two situations.

4.4 The gradient is shared; the level is not

The episode-level correlation between ranking quality and realised reward is almost identical for the two consumers ($r = +0.372$ and $+0.382$; $\rho = +0.392$ and

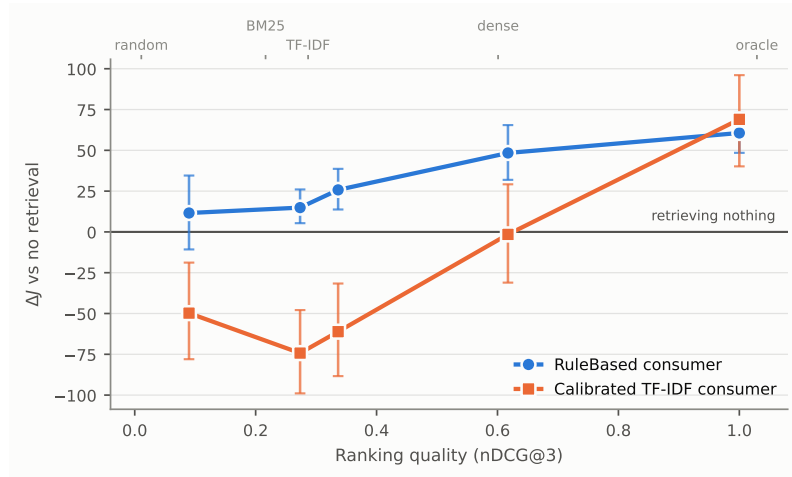


Fig. 1. The same rankings, two consumers, one zero line. The horizontal axis is a property of the retrieval alone; the vertical axis is what that retrieval turned out to be worth, as a paired effect against retrieving nothing (bars are 95% crossed bootstrap intervals). Across the whole practical range the two consumers sit on opposite sides of zero, converging only at the oracle selector. A relevance-only evaluation observes the horizontal axis and cannot distinguish the two situations.

+0.367), computed within episodes so that episode difficulty cannot dominate. This is worth stating precisely, because it narrows the claim. *Within* a consumer, better evidence does produce better decisions, and to a similar degree for both. What differs is the level: the calibrated consumer is worse than retrieving nothing across the whole practical range, and only reaches parity once retrieval is near-oracle. Consumer dependence here is an offset, not a reversed gradient — which is precisely why a relevance-only evaluation cannot detect it.

4.5 More evidence is not better

The last panel of Table 2 follows the strongest ranker across budgets. Moving from $k = 3$ to $k = 5$ raises dense retrieval’s nDCG ($0.617 \rightarrow 0.691$) while lowering its effect on the calibrated consumer ($-1.5 \rightarrow -18.9$) and leaving the rule-based consumer flat ($+48.4 \rightarrow +46.0$). The extra documents are mostly non-relevant, and neither consumer has a mechanism for discounting them — but only one of them pays for that.

5 Locating the mismatch

Why does one consumer convert better evidence into better decisions while the other does not? We hold retrieval aside and vary the belief directly, in two transfer environments: a held-out multi-echelon demand-surge setting (Experiment M)

Table 2. Experiment R. How much of a ranking’s value belongs to the ranking? Left: the retriever $\times$ consumer interaction at $k = 3$ — the difference between the two consumers’ paired effects for the same ranking. Every interaction changes sign, so no single number describes what a ranking is worth. Right: within each consumer, how often ranking quality orders two systems the way realised utility does, and how strongly the two correlate across episodes.

System	RuleBased ΔJ	Calibrated ΔJ	Interaction	Sign flip
Random	+11.6	-49.8	+61.4	✓
BM25	+14.9	-74.3	+89.2	✓
TF-IDF cosine	+25.8	-61.1	+86.9	✓
Dense (MiniLM)	+48.4	-1.5	+49.9	✓

Consumer	Reversal rate	Kendall τ_b	Pearson r	Spearman ρ
RuleBased	0.00 (0/6)	+1.00	+0.372	+0.392
Calibrated TF-IDF	0.33 (2/6)	+0.33	+0.382	+0.367

Dense retrieval at budget k	$k = 1$	$k = 3$	$k = 5$
nDCG@ k	0.556	0.617	0.691
RuleBased ΔJ	+21.3	+48.4	+46.0
Calibrated TF-IDF ΔJ	-69.3	-1.5	-18.9

and a supply-side capacity drop (Experiment S). Alongside the interpreters we run constant-belief endpoints, a no-text operating point tuned only on development seeds, a shuffled-text ablation in which the interpreter reads a real warning belonging to a different template, and a label-only ablation that discards the interpreter’s probabilities.

A text-free control is a demanding baseline. In Experiment M the development-tuned no-text control returns 560.4, above single TF-IDF (516.2), calibrated TF-IDF (530.2) and GPT-4o (534.9); only the rule-based interpreter (568.7) exceeds it. In Experiment S it returns 1725.1 against 1447.0 for the best interpreter. Reward increases monotonically in the assumed event probability, so the controller is systematically under-positioned under the benchmark prior and any belief that raises the order-up-to level captures part of that gap without reading anything. This is the same failure mode that makes the calibrated consumer fragile in Experiment R: its value depends on where its belief mass sits relative to the controller’s operating point, not only on whether the belief is right.

Belief accuracy does not imply operational value. In Experiment S the shuffled-text control returns 1415.2, above the genuine interpreter’s 1372.9, at roughly half the belief accuracy. Whatever that interpreter gains there comes from the shape of its belief distribution, not from identifying the regime. In Experiment M shuffling does destroy the effect (530.2 $\rightarrow$ 451.1, below the uninformed prior), so reading the correct text is doing real work in one environment and not the other.

Table 3. Experiments M (held-out multi-echelon transfer, demand surge) and S (supply-side capacity drop). Rows are grouped as text-free controls, text interpreters, degraded-text controls, and the oracle-belief reference. J_w is the balanced weighted return and ΔJ the paired effect against NoInfo, with crossed bootstrap intervals. The dev-tuned no-text row is the one every interpreter must beat, and in both experiments most do not.

Information source	Experiment M			Experiment S		
	J_w	ΔJ	95% CI	J_w	ΔJ	95% CI
NoInfo (prior)	467.7	+0.0	[+0, +0]	1333.6	+0.0	[+0, +0]
Constant $p_{\text{event}} = 0$	381.2	-86.5	[-93, -80]	1166.7	-166.9	[-167, -167]
Constant $p_{\text{event}} = 0.5$	501.1	+33.3	[+32, +35]	1333.6	+0.0	[+0, +0]
Constant $p_{\text{event}} = 1$	560.4	+92.7	[+90, +96]	1725.1	+391.5	[+387, +396]
No-text (dev-tuned)	560.4	+92.7	[+90, +96]	1725.1	+391.5	[+387, +396]
RuleBased	568.7	+101.0	[+95, +106]	1447.0	+113.4	[+39, +183]
TF-IDF (single)	516.2	+48.5	[+36, +59]	1362.8	+29.2	[-4, +68]
TF-IDF (calib., isotonic)	530.2	+62.5	[+5, +96]	1372.9	+39.3	[-73, +131]
GPT-4o	534.9	+67.1	[+29, +87]	1372.6	+38.9	[-39, +85]
TF-IDF cal. (label only)	523.5	+55.8	[-23, +89]	1392.0	+58.4	[-73, +115]
TF-IDF cal. (shuffled)	451.1	-16.6	[-66, +52]	1415.2	+81.6	[-64, +258]
Perfect-semantic ref.	552.0	+84.3	[+77, +91]	1446.3	+112.7	[+109, +116]

Probabilistic sophistication does not either. Collapsing the calibrated interpreter to its hard label costs little in Experiment M (530.2 $\rightarrow$ 523.5) and helps in Experiment S (1372.9 $\rightarrow$ 1392.0). In the controlled benchmark the label-only ablation of the raw classifier reaches a normalised value of 0.773 against 0.245 for its own probabilities, with a better Brier score (0.222 versus 0.552) and a much worse log-loss (2.558 versus 0.932): the two proper scores disagree about it, and neither predicts its operational advantage.

Calibration is not an isolated intervention. `CalibratedClassifierCV` fits one classifier per fold and averages them, so a raw-versus-calibrated contrast mixes a calibration map with an ensembling change. Adding the missing arm — the same fold ensemble with no calibration map — separates them: ensembling alone *lowers* normalised value (0.245 $\rightarrow$ 0.154) and the calibration map then recovers past it ($\rightarrow$ 0.338). The direction survives; the naive difference is not attributable to calibration.

The perfect-semantic reference is not an upper bound. A correct belief passed through the fixed controller is beaten by a constant belief that reads nothing, in both transfer environments (Table 3). We therefore call it a reference rather than an oracle, and avoid normalising by it where a raw paired effect will do.

Table 4. Experiment B. *Every* one-factor-at-a-time variant, including the adverse ones. Values are the balanced weighted return J_w ; OIV is the oracle information value. $\dagger$ marks a variant whose OIV is not positive: there the oracle-belief reference itself underperforms the uninformed prior, so a normalised ratio cannot be read as recovery of information value and the finding is about controller–environment mismatch, not semantic sensing.

Variant	NoInfo	No-text	Rule	TF-IDF	GPT-4o	Oracle	OIV
baseline	1324	1713	1436	1362	1362	1435	+112
high_noise	1313	1674	1417	1344	1346	1411	+97
long_lead $\dagger$	-876	-967	-902	-869	-863	-878	-3
lost_sales $\dagger$	1512	1439	1489	1503	1507	1490	-22
low_noise	1334	1732	1449	1375	1372	1450	+116
short_lead $\dagger$	2398	2486	2411	2248	2213	2214	-183

6 Boundaries and limitations

Perturbing one environment factor at a time shows how conditional these quantities are. In three of six variants the perfect-semantic reference itself underperforms the uninformed prior, so no normalised information-value statement is meaningful there and we mark rather than report a ratio. Under short lead times the calibrated interpreter returns 2248 and GPT-4o 2213 against the uninformed prior’s 2398: halving the lead time leaves too little time to act, so a belief that shifts stock upward mostly buys holding cost.

The report feed and environments are synthetic, single-SKU and finite-regime. Experiment R’s pools are generated rather than authored, so it carries no language-axis variance; its query is fixed and relevance is defined by construction rather than by assessors, which makes the oracle selector a ceiling for *these* judgments rather than for downstream utility in general. Two consumers do not establish the shape of consumer dependence, only that it is large enough to reverse a sign. The dense retriever is one general-purpose sentence encoder with frozen embeddings, not a trained dense retrieval system. The controller is fixed and not jointly optimised with the interpreter. Cached LLM outputs cover a small set of models with reconstructed rather than certified provenance. Evidence is consumed by concatenation; belief-level pooling over separately interpreted documents is a natural variant we do not evaluate.

7 Conclusion

In a setting where retrieved text drives a decision policy, the value of a ranking is not a property of the ranking. Every retrieval system we tested helps one consumer and harms another, the interaction changes sign in every case, and relevance-based evaluation is a reliable guide for one consumer and unreliable for the other with nothing upstream to distinguish them. The evidence is extractable — an oracle selector helps both — so this is a mismatch problem, not a scarcity

problem. For evaluation practice the implication is narrow and concrete: a retrieval component intended for a decision-making consumer should be scored against that consumer, with a no-retrieval control and a matched budget, and a reported improvement in relevance should not be assumed to survive the handoff.

A Design, accounting and reproduction

Table 5. Experiment accounting, generated from the saved episode files rather than from design intent. ‘Episodes’ counts rows across all information sources; the number of *paired worlds* is seeds $\times$ templates, which is not the episode count. Experiment R draws a generated candidate pool per seed and so has no template axis.

Experiment	Seeds	Templates	Sources	Episodes
R (retrieval)	30	n/a	6	3,240
C (controlled)	20	36	12	8,640
M (multi-echelon)	30	24	12	8,640
S (supply-side)	30	16	12	5,760

No-text operating point. Experiment M selected $p_{\text{event}} = 1.0$ on seeds 3200–3229, Experiment S selected $p_{\text{event}} = 1.0$ on seeds 4200–4229. Both development seed ranges are disjoint from every confirmation seed range, and selection used the balanced estimand. The arm is tuned, but never on its own evaluation data.

Environment, costs and timing. Single-item lost-sales inventory over a finite horizon of $T = 40$ periods: revenue 10 per unit sold, ordering cost 5 fixed plus 2 per unit, holding cost 1 per unit of ending inventory, stockout penalty 8 per unit of unmet demand, initial on-hand 30, Poisson demand with base mean 8. Three regimes: NORMAL; SUPPLIER DELAY, raising the lead time during the event window; and DEMAND SURGE, multiplying the demand rate. The supply-side transfer instead reduces a factory’s capacity in a multi-echelon network. Within a period the order is: receive arrivals, realise demand, fulfil from on-hand, place an order. An order placed in period t under lead time L arrives at the start of period $t + L$; orders already in transit keep their arrival period, so a lead-time change applies only to orders placed from that period onward. The simulator and every planner share one implementation of this schedule.

Information release. Warning text is published at period 12; the event window begins at period 20. Before publication no interpreter has read anything and every controller plans under the benchmark prior. From publication the interpreted belief replaces the prior while on-hand stock and the in-transit pipeline carry over.

Belief to action. The controlled experiments solve a receding-horizon linear program each period and execute only its first order, with belief-weighted expected lead time and demand per future period. The multi-echelon experiments use a belief-adaptive base-stock policy with effective rate $\mathbb{E}_b[\mu]$ and order-up-to target $\mathbb{E}_b[\mu]\bar{L} + 1.65\sqrt{\bar{L}\mathbb{E}_b[\mu]}$. The controller is identical across information sources within an experiment.

Interpreters, information and retrieval. No non-oracle interpreter receives the regime, future demand, future state or environment internals. The TF-IDF classifiers are fitted only on training templates and evaluated on disjoint held-out ones. LLM beliefs are replayed from frozen canonical responses, with model labels reconstructed by re-deriving each cache key from every known (model, text) pair and unmatched entries labelled `unknown`. BM25 uses the Okapi form with $k_1 = 1.5$, $b = 0.75$ and idf floored at zero; dense retrieval is cosine similarity over frozen `all-MiniLM-L6-v2` embeddings, shipped so the published numbers reproduce without model weights. Ties break by document id.

Reproduction. The release reconstructs its own offline inputs: cached semantic responses come from a tracked, checksum-verified manifest, and classifier checkpoints are retrained from tracked templates with fixed seeds; neither step needs credentials or network access. The multi-echelon simulator is a pinned published package [1], and every stored episode replays through it to its recorded reward exactly. A verification entry point regenerates every table in this paper from the stored episodes and fails if any differs. This paper shows that a decision-support component can look better on upstream metrics while being worse to act on, so deployment needs human oversight and conservative fallbacks; a synthetic study does not establish deployment safety.

References

1. gym-invmgmt: a multi-echelon inventory management Gymnasium environment with configurable topology and composable demand engines. Python package (2026), version 0.2.1. Source distribution `gym_invmgmt-0.2.1.tar.gz`, SHA-256 `a2a286cd95d9967b27db5ed5301d262862dd98bfbe1b50358fc9944af1d60311`
2. Borlund, P.: The concept of relevance in IR. *Journal of the American Society for Information Science and Technology* **54**(10), 913–925 (2003). <https://doi.org/10.1002/asi.10286>
3. CUE-R authors: CUE-R: Intervention-based per-evidence utility estimation for retrieval-augmented generation. <https://arxiv.org/abs/2604.05467> (2026), arXiv preprint arXiv:2604.05467; preprint, not peer reviewed
4. DeGroot, M.H., Fienberg, S.E.: The comparison and evaluation of forecasters. *The Statistician* **32**(1), 12–22 (1983). <https://doi.org/10.2307/2987588>
5. Donti, P.L., Amos, B., Kolter, J.Z.: Task-based end-to-end model learning in stochastic optimization. In: *Advances in Neural Information Processing Systems*. vol. 30 (2017), <https://papers.neurips.cc/paper/2017/hash/3fc2c60b5782f641f76bcefc39fb2392-Abstract.html>

6. Elmachtoub, A.N., Grigas, P.: Smart “predict, then optimize”. *Management Science* **68**(1), 9–26 (2022). <https://doi.org/10.1287/mnsc.2020.3922>
7. Farahmand, A.M., Barreto, A., Nikovski, D.: Value-aware loss function for model-based reinforcement learning. In: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research*, vol. 54, pp. 1486–1494. PMLR (2017), <https://proceedings.mlr.press/v54/farahmand17a.html>
8. GroGU authors: GroGU: Model-specific grounding utility for retrieval. <https://arxiv.org/abs/2601.23129> (2026), arXiv preprint arXiv:2601.23129; preprint, not peer reviewed
9. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: *Proceedings of the 34th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 70, pp. 1321–1330. PMLR (2017), <https://proceedings.mlr.press/v70/guo17a.html>
10. Howard, R.A.: Information value theory. *IEEE Transactions on Systems Science and Cybernetics* **2**(1), 22–26 (1966)
11. Hubbs, C.D., Perez, H.D., Sarwar, O., Sahinidis, N.V., Grossmann, I.E., Wassick, J.M.: Or-gym: A reinforcement learning library for operations research problems (2020), <https://arxiv.org/abs/2008.06319>
12. Järvelin, K., Kekäläinen, J.: Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems* **20**(4), 422–446 (2002)
13. Lambert, N., Amos, B., Yadan, O., Calandra, R.: Objective mismatch in model-based reinforcement learning. In: *Proceedings of the 2nd Conference on Learning for Dynamics and Control. Proceedings of Machine Learning Research*, vol. 120, pp. 761–770. PMLR (2020), <https://proceedings.mlr.press/v120/lambert20a.html>
14. Mandi, J., Bucarey, V., Tchomba, M.M.K., Guns, T.: Decision-focused learning: Through the lens of learning to rank. In: *Proceedings of the 39th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 162, pp. 14935–14947. PMLR (2022), <https://proceedings.mlr.press/v162/mandi22a.html>
15. Niculescu-Mizil, A., Caruana, R.: Predicting good probabilities with supervised learning. In: *Proceedings of the 22nd International Conference on Machine Learning*. pp. 625–632 (2005)
16. Reimers, N., Gurevych, I.: Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. pp. 3982–3992 (2019)
17. Robertson, S.E., Walker, S.: Some simple effective approximations to the 2-Poisson model for probabilistic weighted retrieval. In: *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. pp. 232–241 (1994)
18. Salemi, A., Zamani, H.: Evaluating retrieval quality in retrieval-augmented generation. In: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. pp. 2395–2400. ACM (2024). <https://doi.org/10.1145/3626772.3657957>
19. Zhao, S., Kim, M., Sahoo, R., Ma, T., Ermon, S.: Calibrating predictions to decisions: A novel approach to multi-class calibration. In: *Advances in Neural Information Processing Systems*. vol. 34, pp. 22313–22324 (2021), https://proceedings.neurips.cc/paper_files/paper/2021/hash/bbc92a647199b832ec90d7cf57074e9e-Abstract.html